Abhinav Jagan Polimera

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Education

Birla Institute of Science and Technology, Pilani - Hyderabad Campus **2020-2024**

- B.E Computer Science - 8.2/10 CGPA

Work Experience

Cisco Systems **Apr 2026 -Present**

Software Developer 2, Distributed Systems organizationn *On-site*

- **Designed** a next-generation ERSPAN solution by introducing **GUE (UDP-based) encapsulation** as a scalable alternative to traditional GRE, **leading end-to-end feature development** as part of customer-driven feature to **Microsoft** to enable efficient packet mirroring with **reduced overhead**, achieving hardware-accelerated **performance optimization**.
- Contributed to an **AI-driven Infrastructure-as-a-Service** initiative for deploying **quantized ML models** on **Cisco routers**; applied **data-driven model selection** on large-scale network telemetry to identify patterns and choose optimal ML approaches; **delivered concept-to-prototype with a 4-member core team**; **recognized across Cisco AI forums** and currently in **customer evaluation phase for deployment**.

Cisco Systems **Aug 2024 -Apr 2026**

Software Developer, Distributed Systems organizationn *On-site*

- Led **end-to-end delivery** of a traffic mirroring feature that involves enabling **mirroring of egress traffic within the router network** for **Cisco 8000 (Silicon One) routers**, achieving **functional parity** with NCS 5000 routers for SoftBank's migration, and leveraged **component design expertise** and **AI-assisted development** to reduce delivery time by **60 %**.
- Owned **end-to-end delivery, debugging, and POC ownership** for **traffic mirroring features (SPAN, Lawful Intercept, NetFlow)** on **Cisco 8000 (Silicon One)** across Cisco's OS stack, driving **cross-functional execution** and leading **multi-platform topology bring-up** for upcoming hardware.
- **Spearheaded** the deployment of a scalable automation infrastructure using open-source and Cisco internal frameworks, enabling seamless **sanity testing**, **faster debugging**, and improved **product quality**; independently developed **AI-powered productivity tools**, including a **metrics analyzer for leadership insights** and a unified **traffic mirroring platform** featuring **support matrix validation, drag-and-drop router simulation (LCs, RPs, topology, traffic generators)**, and **MCP-based modular components** for enhanced usability.

Cisco Systems **Jan 2024 - June 2024**

Technical Intern, Distributed Systems organization *On-site*

- Optimized router OS feature testing and deployment through test design refinement and parallel execution; initiative earned an internal Cisco excellence award.
- Led end-to-end automation lifecycle with Python for NTP VRF OpenConfig feature delivery.

Shris Infotech **June 2022 - Jul 2022**

Software Intern *Remote*

- Developed a one-click face-recognition calling app using Python (DeepFace, OpenCV) with Flutter and Dart.

Projects

Playlistify - web development **-**

- **Identified a product usability gap in Spotify** and **independently built Playlistify**, a **Python- and Flask-based web app leveraging the Spotify Web API**, that **converts liked songs into shareable playlists with a single click**, demonstrating **strong problem-solving initiative, curiosity, and full-stack development skills**.

Remote Drone surveillance- Machine learning/ image processing **Publication Link**

- **Researched and implemented** an automated **drone-based surveillance system** using **Python, OpenCV, YOLO, and image processing** to **detect and classify objects in real time**; designed **threshold-based algorithms** using **object count** and **activity duration** to **automatically flag illegal actions**, strengthening **remote security threat detection**.

Performance Assessment of Neural Radiance Fields and Photogrammetry **Publication Link**

- **Researched and implemented** a comparative study between **Neural Radiance Fields (NeRF)** and **traditional Photogrammetry** for **3D reconstruction** of **man-made and natural features**, leveraging **neural networks** and **volumetric rendering** in **Python**; the work was **selected and presented at the ICAART conference**.